

Algal Growth Bulletin

A four-page monitoring circular

Summary

Growth rate of *Chlorella vulgaris* was measured in samples B1-B4 under a six-day light cycle. Optical density was compared with a blank control before the logarithmic estimate was calculated.

Sampling

Bottles were mixed by inversion and read at 680 nm. The blank control was measured twice because condensation appeared on the first cuvette.

$$\mu = (\ln N_2 - \ln N_1) / (t_2 - t_1)$$

Methods note

The optical density series was corrected against the blank control at each day. A bottle was retained when duplicate readings differed by less than 0.012 absorbance units.

The optical density record was compared with the blank control for each day. A bottle remained eligible when the paired readings differed by less than 0.012 absorbance units.

One archived line was abraded during handling; the visible sequence reads: B3 exposure was re... at Day 4 before incubation resumed.

Sample	Day 0	Day 2	Day 4	Day 6
B1	0.083	0.141	0.244	0.398
B2	0.081	0.137	0.230	0.376
B3	0.086	0.129	0.201	0.312
B4	0.079	0.084	0.087	0.090

Results

B1 had the largest growth rate, while B4 remained close to the blank control. The B3 interruption prevents a continuous Day 2-Day 4 interpretation but does not erase the recorded values.

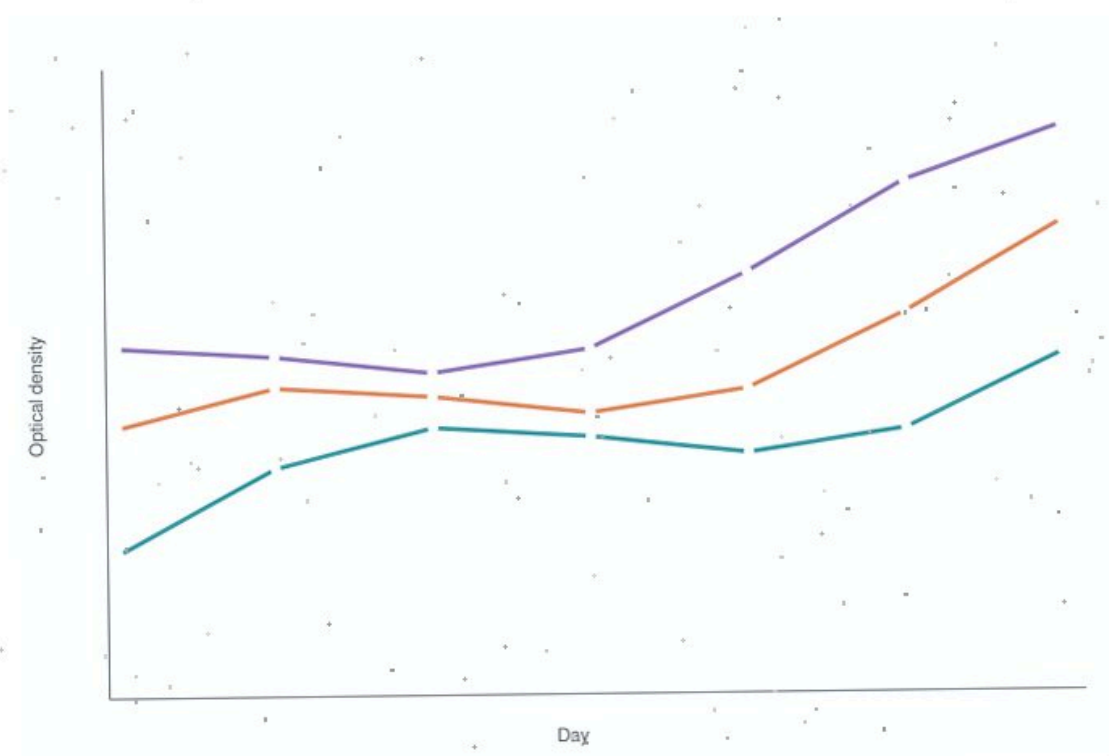


Figure 1. Optical density trajectories for B1-B4. The lowest trace includes the blank control range.

CHECK B4

Interpretation

The bulletin supports a difference between actively growing bottles and the blank-like B4 series. It does not establish why B4 changed slowly or whether the response persists under field light:

Footnote 1. Day labels refer to elapsed incubation time, not calendar dates.

References

- [1] Hale T. Blank correction in small-volume algal assays. *Freshwater Methods* 9:33-41.
- [2] Ibrahim N. Reporting interrupted growth series. *Monitoring Circular* 12:5-11.
- [3] Voss P. Optical density limits in mixed cultures. *Lake Analytics* 3:77-85.